

DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Programme: B.Tech. Semester IV
Paper Code: MAIC-204
Subject: Applied Linear Algebra
Date: 21-02-2026

Examination: Mid Term I
Paper: Regular and Reappear
M.M.: 15
Time: 11.00 a.m.-11.50 a.m.

Note: Attempt all the questions.

1. If w_1, w_2, w_3 are linearly independent vectors, show that the sums $v_1 = w_2 + w_3, v_2 = w_1 + w_3, v_3 = w_1 + w_2$ are linearly independent. [4]
2. Find the linear operator T on $\mathbf{R}^2$ such that $T(1, -1) = (1, 0); T(2, -1) = (0, 1)$ and $T(-3, 2) = (1, 1)$. [4]
3. Let T be the linear operator on $\mathbf{R}^3$ defined by
 $T(x, y, z) = (3x, x - y, 2x + y + z)$. Is T invertible? If so, find a rule T^{-1} like the one that defines T . [4]
4. Let V be the vector space spanned by finite set of vectors $\beta_1, \beta_2, \dots, \beta_m$. Prove that any independent set of vectors in V is finite and contains no more than m elements. [3]